

REVIEW

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AI-based live birth prediction in IVF cycles: a systematic review without meta-analysis of model performance and validation

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Abstract

Background Artificial intelligence and machine learning have been identified as promising avenues for improving embryo selection and predicting the outcome of in vitro fertilization (IVF). However, there is much heterogeneity in terms of model architectures and validation methodologies used for these models, as well as their actual clinical utility in terms of the end outcome of achieving a live birth.

Objective To evaluate the diagnostic performance and validation methodologies of artificial intelligence-based predictive models for the end outcome of achieving a live birth following in vitro fertilization cycles.

Methods This systematic review without meta-analysis was carried out following the PRISMA 2020 guidelines, and the study was registered on the PROSPERO database (CRD420261298569). An extensive literature search was carried out on PubMed/MEDLINE, Scopus, EMBASE, Web of Science, and CENTRAL databases for original studies published between January 2010 and January 2026. Original studies on the development or validation of AI models for embryo evaluation or prediction of IVF outcomes, where live birth was the primary or secondary outcome, were considered for inclusion. The methodological study quality and risk of bias assessment were done using the QUADAS-2 tool, while the certainty of the evidence was assessed using the modified GRADE approach.

Results A total of 23 primary clinical studies (20 retrospective cohort studies, 2 prospective cohort studies, and 1 randomized controlled trial) were included. In total, model evaluation on the validation cohorts comprised around 45,000 embryos/cycles, while the training datasets exceeded 200,000 embryos/cycles. Diagnostic accuracy was highly variable depending on model architecture. Multimodal models combining embryo image data and patient-level clinical variables demonstrated the highest model performance. These models achieved up to 0.97 Area Under the Curve (AUC) values and accuracy ranging from 74% to 82%. In contrast, Time-Lapse models achieved 0.64 to 0.97 AUC values (64% to 78% accuracy), while models using clinical variables alone achieved 0.70 to 0.80 AUC values (76% to 78% accuracy), and models using static images demonstrated the lowest performance (62% to 69% accuracy). Center-based models always outperformed national registry-based models. It is also interesting to note that the

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single prospective, double-blind randomized controlled trial evaluating an AI algorithm (iDAScore) against traditional morphology did not demonstrate noninferiority with clinical pregnancy rates of 46.5% vs. 48.2%, respectively.

Conclusions In terms of the diagnostic potential of the AI-based predictive models for the prediction of live birth following IVF, these are highly effective when utilizing multimodal prediction approaches to assess embryonic competence and the systemic reproductive environment. The existing body of evidence is heavily based on retrospective cohorts and is subject to a number of limitations due to the presence of risks for selection bias and internal validation overfitting. As prospective RCT data have not been used to prove the independent clinical superiority or non-inferiority of AI-based prediction for IVF outcomes, currently, it should be used as a supplementary tool for embryologists only.

Keywords Artificial intelligence, Machine learning, Embryo selection, In vitro fertilization (IVF), Live birth prediction, Time-lapse imaging, Clinical pregnancy outcomes, Systematic review

Introduction

In vitro fertilization (IVF) has transformed reproductive medicine, and it has given infertile couples a chance for biological parenthood. However, despite significant improvements in ovarian stimulation protocols, laboratory techniques, and embryo transfer methodologies, the success rates are still low, and live birth rates are generally 20% to 50%, depending on the maternal age and other clinical factors [1]. One of the most important bottlenecks in the IVF sequence of procedures is the process of selecting viable embryos for transfer. Approximately 40% of morphologically “normal” embryos do not implant after transfer, apparently indicating that traditional methods of morphological assessment are not completely indicative of embryonic competence [2].

The current methods of embryo evaluation are mostly based on visual microscopic evaluation of morphological characteristics at certain stages of development, most commonly at the blastocyst stage (day 5 or 6). Embryologists usually use the Gardner and Schoolcraft classification for grading blastocysts, which evaluates blastocoele expansion and also provides separate grades for the ICM and TE as A, B, and C according to their size, compactness, and cohesiveness (Gardner & Schoolcraft, 1999). Even though this grading system is commonly used and is clinically relevant, it is subjective and is associated with a high degree of inter- and intra-observer variability for both cleavage stage and blastocyst stage embryo evaluation, even for experienced embryologists [3, 4]. While this approach is widely adopted, it is subjectively based and shows a considerable amount of interobserver and intraobserver variability, even in the hands of experienced embryologists [4]. This limitation has stimulated research on methods to provide objective, standardised methods for embryo assessment, such as time-lapse monitoring (TLM) of embryonic development and morphokinetic analysis [5].

The advent of artificial intelligence, specifically the deep learning and machine learning techniques, has provided new opportunities for advancing embryo selection. These

types of computational approaches have the potential to process large volumes of image data and clinical variables to determine complex patterns of reproductive success predictive of reproductive success [6]. AI-based models have been trained with thousands to hundreds of thousands of embryo images and their pregnancy outcomes, which may allow more objective and thorough evaluation of embryonic viability [7]. Several commercially available AI tools have been created and implemented into the clinical environment worldwide, with varying degrees of supporting clinical evidence.

Despite the recent interest in AI-based embryo assessment, there are major gaps in the evidence on the true clinical utility and generalisability of these models. Key challenges include: (1) notable heterogeneity in model architecture, training data sets and input variables across studies; (2) less information available from prospective, randomised controlled trial evidence regarding the performance of AI-based selection in comparison with standard morphological assessment; (3) a lack of consistent reporting of diagnostic accuracy metrics and outcome definitions; (4) questions of the generalisability of models trained in single centre or limited multi-centre datasets to varied clinical populations and laboratory settings; and (5) a “blackbox” behaviour of the algorithms in the Deep learning techniques, which limits the transparency and clinical interpretation [8].

Apart from clinical and laboratory heterogeneity, the assessment of AI-based embryo selection systems is also a complex task. In fact, as emphasized by Kragh and Karstoft (2021) [9], AI systems can be optimized for ranking embryos within a cohort or for predicting absolute probabilities of pregnancy or live birth, and it is common for studies to combine these objectives and present non-comparable metrics such as accuracy, AUC, or ranking metrics for different definitions of outcomes and data foundations.” In fact, the authors also demonstrated the potential for “selection bias in retrospective comparisons of AI systems and human embryologists when only transferred embryos are analyzed.

Similarly, alongside the diagnostic accuracy tools QUADAS-2 and bias assessment tools, the revised TRIPOD + AI statement also offers consolidated reporting recommendations for prediction models developed by either regression or machine learning approaches, while also addressing fairness, calibration, and external validation of clinical prediction models [10]. These recommendations are highly relevant to future AI-based embryo prediction studies and guided the assessment of the reporting quality of the studies under review.

While previous systematic reviews have been conducted into applications of AI in embryo assessment [11–13], a systematic review to address this gap by providing a comprehensive qualitative overview of the predictive models for live birth, their diagnostic performance characteristics, validation methodologies, and generalizability is still lacking. Most available reviews have combined results from heterogeneous outcome definitions (including implantation, clinical pregnancy, and live birth) and have not assessed study quality and risk of bias systematically and with the use of standardized diagnostic accuracy assessment tools.

This systematic review synthesizes evidence up to January 2025 to respond to outstanding questions: (1) diagnostic performance of AI-based models in predicting live birth outcomes compared with standard assessment; (2) comparative performance of models of different architectures and input modalities; (3) quality of prospective validation studies; (4) mechanistic understanding of model predictions; and (5) clinical implications for the evidence-based implementation of these models in fertility clinical practice [14].

Methods

Protocol and registration

This systematic review was undertaken according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidelines [15]. The review protocol was developed prospectively, before the extraction of the data and assessment of bias in the extracted data. The review was undertaken in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocol (PRISMA-P) guidelines for transparency and methodological rigor. The review protocol was registered in PROSPERO with a registration number of CRD420261298569 on 1st February 2025 [16], after the completion of the initial search strategy but before extraction of the data, assessment of bias in the extracted data, and synthesis of the extracted data. This is recognized as a limitation of registering a review protocol but is in line with the PROSPERO guidelines for registering review protocols [17].

Information sources and search strategy

A comprehensive literature search was conducted across five major biomedical databases: - PubMed/MEDLINE (National Library of Medicine) - Scopus (Elsevier) - EMBASE (Ovid) - Web of Science (Clarivate Analytics) - Cochrane Central Register of Controlled Trials (CENTRAL).

All searches were carried out with no language restriction. The search strategy was developed and revised with the information of medical research librarians and subject matter experts in reproductive medicine.

Search Timeframe: Database January 2010 to January 2026.

The comprehensive time period was chosen to represent the entire trajectory of the AI-based embryo assessment research from the proof-of-concept research to recent multicenter validation trials.

Complete search strategies

Search terms (Boolean Operators)

The search was designed around four key concepts: artificial intelligence/machine learning, IVF/assisted reproduction, embryo selection/assessment, and clinical outcomes with live birth as the primary outcome of interest. The database-specific syntax was modified accordingly for each database by using controlled vocabulary and field tags. This included the use of MeSH terms for PubMed/Medline and Emtree terms for Embase, as well as free-text words.

Search terms

The search strategy combined the following concepts using Boolean operators:

“artificial intelligence” OR “machine learning” OR “deep learning” OR “neural network” OR “convolutional neural network” OR “random forest” OR “support vector machine” OR “ensemble learning”

AND.

“in vitro fertilization” OR “IVF” OR “assisted reproductive technology” OR “ART” OR “embryo transfer”

AND.

“embryo selection” OR “embryo assessment” OR “blastocyst” OR “embryo grading” OR “embryo morphology” OR “embryo quality”

AND.

“live birth” OR “clinical pregnancy” OR “implantation” OR “pregnancy outcome” OR “reproductive success”

Database adaptation

For PubMed/MEDLINE, MeSH words were employed if available; for EMBASE, the search was modified accordingly with Emtree words. To maximize recall, additional free-text words were added in the updated search process. These words included specific words related to each

study identified by recent embryo-AI publications, such as “live birth prediction,” “iDAScore,” “KIDScore,” “annotation-free embryo scoring,” and “center-specific models.”

Study selection: inclusion and exclusion criteria

Inclusion criteria

Study design: Original research articles describing the development and/or validation of AI or ML models for embryo assessment or IVF outcome prediction.

Population: Patients undergoing IVF/ICSI treatment with embryo transfer.

Intervention: AI models such as deep learning models, machine learning models, neural networks, and related predictive models for embryo assessment or IVF outcome prediction.

Comparator: Standard morphological evaluation, other AI models, or embryologist evaluation for diagnostic accuracy.

Outcomes: The primary outcome of interest was live birth.

Other outcomes of interest were clinical pregnancy or implantation; however, these were eligible for inclusion only if they were linked to embryo selection model performance and live birth was reported as a primary, secondary, or follow-up outcome.

Publication type: Peer-reviewed publications in the English language.

Data availability: Articles were eligible for inclusion if they reported quantitative performance metrics such as sensitivity, specificity, accuracy, or area under the curve of the receiver operating characteristic curve or other quantitative diagnostic/predictive metrics.

Exclusion criteria

(1) Non-original publications, reviews, commentaries, editorials, opinion pieces, conference abstracts without full text, and other forms of secondary research articles were excluded from the primary set; (2) studies of non-embryological AI applications in IVF, such as oocyte selection, sperm analysis, uterine assessment, and other forms of non-embryo-related reproductive tasks, were excluded unless they directly predicted embryo-related live births or pregnancies; (3) animal studies, in vitro studies, simulation studies, and those lacking clinical outcome data were excluded; (4) studies of traditional morphokinetics and morphologic assessment without an AI or ML component were excluded; (5) case reports and case series were excluded; (6) those not reporting any form of quantitative predictive/diagnostic performance metrics were excluded; and (7) in the event that multiple publications were published describing the same cohort or model, the most recent and comprehensive report was included.

For transparency, studies with clinical pregnancy or implantation as primary outcomes were included only when relevant to embryo selection model evaluation for predicting live birth; reviews and non-original studies were used only for background information and not included as primary studies.

Study selection process

Two independent reviewers used standardised screening forms piloted on 50 randomly selected records to screen all titles and abstracts against preset eligibility criteria [18]. Two review authors independently evaluated full-text articles by predefined criteria [19]. Disagreements were resolved through discussion and consensus, and when consensus could not be reached, a third independent reviewer provided adjudication [20]. Inter-rater reliability was calculated with the use of Cohen's kappa statistic [21].

Data extraction

A standardized, comprehensive data extraction form was created by pilot testing in five studies with diverse methodologies [22]. All team members participated in an iterative fashion in refining the instrument [23].

Extracted data elements

1. Study Characteristics: Author, year of publication, country, study design, time of follow-up, and funding source. 2. Population Characteristics: Sample size (embryos and cycles), maternal age, BMI, etiology of infertility, indications of IVF. 3. Model Development: Architecture of AI (CNN, LSTM, ensemble, etc.), type of data input (static images, time-lapse, clinical variables, genetic data), size and composition of training data set, number of centers providing data. 4. Validation Approach: Internal validation approach (k-fold cross-validation, holdout test set), external validation approach (prospective and independent cohort), multicenter validation approach. 5. Outcome Definitions Primary and secondary outcomes Timing of live birth ascertainment Outcome ascertainment methodology 6. Diagnostic Accuracy Metrics Sensitivity Specificity Accuracy Area under the receiver operating characteristic curve (AUC) Positive predictive value Negative predictive value Likelihood ratios, if reported 7. Comparator Performance: Accuracy of standard morphological assessment/Other comparative methods. 8. Model Performance Reporting: Calibration metrics, generalizability studies, and subgroup studies by age or other factors. 9. Study Quality and Risk of Bias Domains evaluated using QUADAS-2 tool (see Risk of Bias Assessment section).

Data were managed using a secure electronic spreadsheet and exported to qualitative summary tables.

Quantitative data handling and synthesis

This review was designed and carried out as a systematic review without meta-analysis. Performance metrics such as accuracy, sensitivity, specificity, and AUC values were thus described descriptively instead of being pooled quantitatively. As described above, for each clinical study, we extracted the reported performance metrics as provided in the original publications. When studies were reporting specific percentages or measures of performance, these were taken directly from the individual trials or cohorts and are accompanied by the relevant study reference. Where ranges (e.g., 60–81%) are reported for a model category, they refer to the minimum and maximum values reported in that model category across all contributing studies, rather than to the mean of those values. Due to the large variability in the study design, model architectures, input data modalities, validation approaches, and outcome definitions, we did not perform formal meta-analysis nor conduct cross-study statistical testing, and all the quantitative comparisons should be considered summary descriptions of individual study results.

Risk of bias and quality assessment

The revised Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool was used to evaluate risk of bias and applicability concerns of included diagnostic accuracy studies [24]. QUADAS-2 assesses four main domains of diagnostic studies:

Patient Selection: - Risk of Bias Questions: Were patients recruited consecutively or randomly? Were the inclusion/exclusion criteria well defined and justified? Was there avoidance of spectrum bias (appropriate case mix)? - Applicability Concern: Does the population studied represent the target population (women undergoing IVF)?

Index Test (AI Model): - Risk of Bias Questions: Was the index test (AI algorithm) interpreted without knowledge of reference standard results (blinding)? Were diagnostic threshold criteria based a priori? Applicability Concern: Are the index test methods adequately described and reproducible?

Reference Standard (Clinical Outcome): - Risk of Bias Questions: Was the reference standard (live birth, clinical pregnancy) appropriate and consistently applied? Great - assessors blinded with index test results? - Applicability Concern: Does the reference standard accurately measure the outcome of interest? Each domain was categorized as Low Risk, High Risk, or Unclear Risk based on the published QUADAS-2 guidance [24]. Overall study quality using descriptive statistics for risk categories

Certainty of evidence assessment

The GRADE (Grading of Recommendations Assessment, Development and Evaluation) methodology was modified to determine the certainty of evidence for primary outcomes by category of timing [15, 25]. Evidence is rated as high, moderate, low, and very low certainty based on study design limitations, inconsistency, indirectness, imprecision, and publication bias considerations.

Results

Study selection

The total database search produced 2,862 unique records. After the title and abstract screening, a total of 316 potentially eligible articles were retrieved for full-text. Following a detailed full-text review, 23 studies satisfied the inclusion criteria and were considered for inclusion in the systematic review. High rates of inter-rater agreement (Cohen's kappa, title/abstract = 0.84 and full text, kappa = 0.79) (Fig. 1).

Study characteristics and geographic distribution

Among the 23 included primary clinical studies, 12 studies (52%) were conducted in Asia, 5 studies (22%) in Europe, 2 studies (9%) in North America, 3 studies (13%) involved multicentre international collaborations, and 1 study (4%) was conducted in Australia. Publication years ranged from 2019 to 2026. Table 1 summarizes key study characteristics, including study design, sample size, AI model type, input data modality, timing category, and primary outcome.

Study design distribution

The 23 clinical studies included were of different types: 20 were retrospective cohort studies (87%), 2 were prospective cohort studies (9%), and 1 was a multicenter randomized controlled trial (4%). The total number of embryos/cycles for model evaluation across clinical validation cohorts was around 45,000, while the datasets for training were over 200,000 embryos/cycles from large time-lapse and registry datasets (Fig. 2).

Primary efficacy outcomes by model type

Time-lapse based models

The individual accuracy of 11 clinical studies in which the AI model is trained on time-lapse imaging data varied from 64% to 78%, and the AUC varied from 0.64 to 0.97. Huang et al. (2022) [26] have reported an AUC of 0.968 for the prediction of live birth by the ResNet-like CNN model in a large cohort of 15,434 single blastocyst transfers. Ueno et al. (2022) [33] reported an AUC of approximately 0.70 for live birth prediction using the annotation-free iDAScore v1.0 across 3,010 single vitrified-warmed blastocyst transfers. Similarly, Lee et al. (2024) [27] reported an AUC of 0.64 for live birth

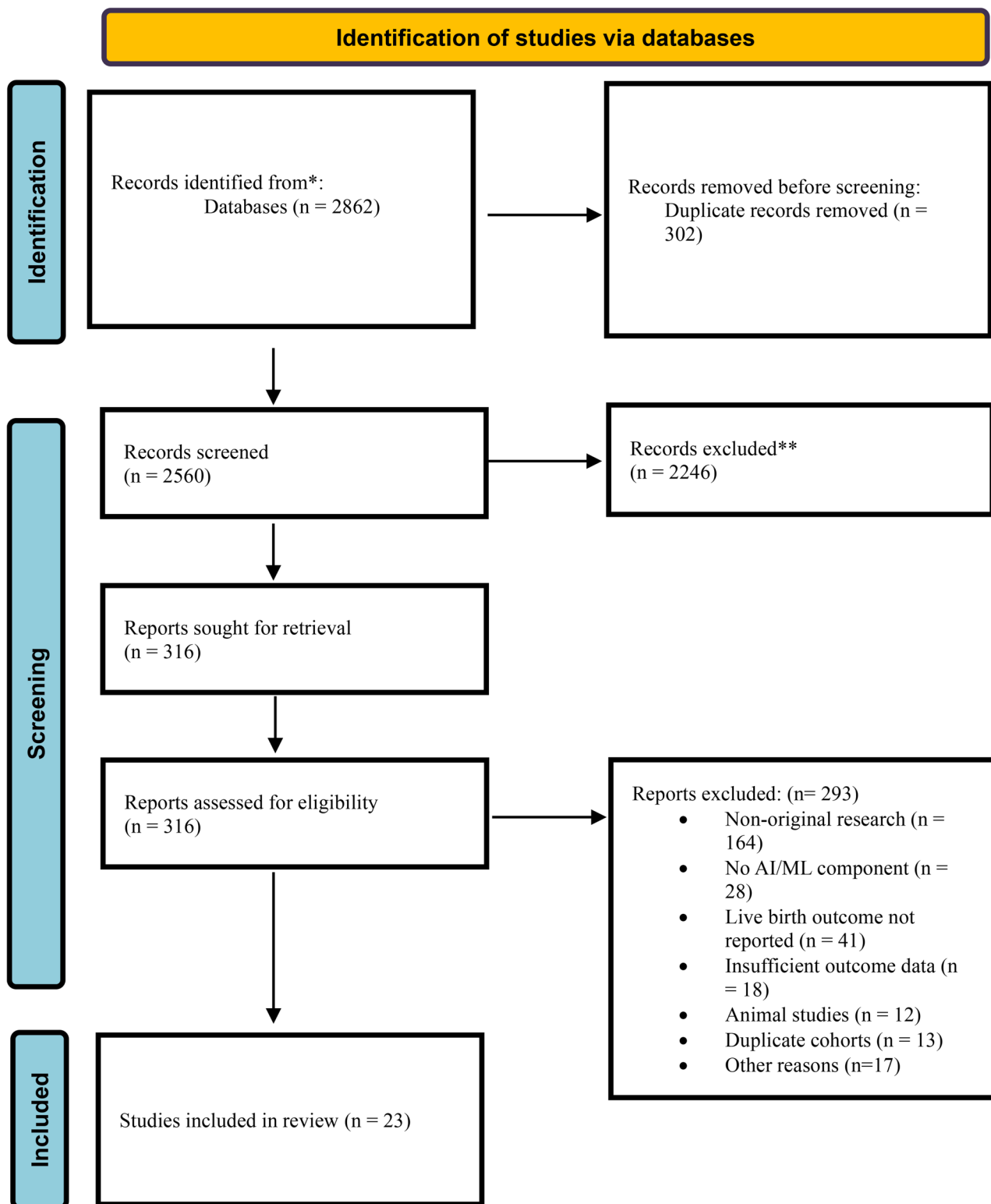
**Fig. 1** PRISMA 2020 Flow Diagram

Table 1 Characteristics of Included Studies (n = 23)

Author, Year	Country	Study Design	Sample (Embryos/Cycles)	Model Type	Input Data	Primary Outcome
Huang et al., 2022 [26]	China	Retrospective single-centre cohort	15,434 single blastocyst transfers	Deep learning (ResNet-like CNN) on time-lapse blastocyst images.	Time-lapse videos (blastocyst frames) only	Live birth per embryo transfer
Lee et al., 2024 [27]	Taiwan	Retrospective single-centre cohort	482 SET cycles after PGT-A	iDAScore-based AI scoring (v1.0)	Time-lapse sequences scored by iDAScore, plus PGT-A results	Association between AI score quartiles and live birth probability; AUC for live-birth prediction 0.64
Papamentzelopoulou et al., 2024 [28]	Greece	Retrospective monocentric cohort	429 blastocysts from 91 IVF/ICSI cycles	Comparison of KIDScore D5 (manual morphokinetic) vs. iDAScore (AI)	Time-lapse images with manual annotations and automated iDAScore	Live birth; ROC AUC 0.695 for KID5 and 0.657 for iDAScore
Nguyen et al., 2024 [29]	Multicentre, USA	Multicentre retrospective	Multiple US fertility centres	Gradient-boosting/ensemble ML models trained on centre-specific data vs. registry-based model	Clinical and demographic variables	Live birth, reported as live-birth probabilities per cycle
Bereczki et al. 2025 [2]	Hungary	Retrospective cohort	8,142 cycles	ML algorithms (XGB, RF, SVM)	Clinical variables (age, AMH, BMI)	Live birth, Clinical pregnancy
Yao et al. 2025 [30]	USA	Retrospective cohort	18,500 cycles (multi-center)	Center-specific ML models	Clinical preprocedural variables	Live birth per cycle
Illingworth et al. 2024 [31]	Australia/Europe	RCT, Double-blind	1,066 patients	Deep learning (iDAScore)	Time-lapse videos	Clinical pregnancy rate
Liao et al. 2021 [5]	China	Retrospective cohort	1,400 embryos	Deep learning (CNN)	Time-lapse images	Blastocyst formation quality
Sawada et al. 2021 [6]	Japan	Retrospective cohort	558 embryos	AI algorithm	Time-lapse images	Live birth
Enatsu et al. 2022 [7]	Japan	Retrospective cohort	285 embryos	AI algorithm with visualization	Time-lapse images	Blastocyst viability
Khosravi et al. 2019 [8]	Multi-center	Retrospective cohort	8,886 embryos (11 centers)	Deep learning (CNN)	Static light microscope images	Clinical pregnancy outcome
He et al. 2024 [11]	China	Retrospective cohort	1,200 embryos	AI algorithm	Time-lapse images	Embryo ploidy, Implantation
Dehghan et al. 2024 [12]	Iran	Retrospective cohort	500 IVF cycles	ML + genetic algorithm	Clinical + embryo quality	IVF success
Blank et al. 2019 [13]	Belgium	Retrospective cohort	1,500+ embryos	Machine learning	Morphological features	Implantation prediction
Miyagi et al. 2019 [14]	Japan	Retrospective cohort	560 embryos	AI applied to blastocyst images	Blastocyst images	Live birth prediction
Coticchio et al. 2021 [32]	Italy	Retrospective cohort	812 embryos	AI algorithm	Time-lapse (cytoplasmic movements)	Blastocyst development
Hojjati et al. 2026 [18]	Iran	Retrospective cohort	1,200 embryos	3D-CNN, hybrid, transfer learning	3D embryo images	Blastocyst selection
Theilgaard-Lassen et al. 2023 [19]	Denmark	Retrospective cohort	181,428 embryos (multi-center)	Deep learning (iDAScore v2.0)	Time-lapse video sequences	Live birth by transfer day
Kalyani et al. 2024 [20]	India	Retrospective cohort	1,800 embryos	Deep learning (CNN)	Time-lapse images	Blastocyst formation and quality
Tran et al. 2019 [21]	Australia	Retrospective cohort	1,140 embryos	Deep learning	Time-lapse incubation images	Fetal heart pregnancy

Table 1 (continued)

Author, Year	Country	Study Design	Sample (Embryos/Cycles)	Model Type	Input Data	Primary Outcome
Ueno et al., 2022 [33]	Japan	Large single-centre retrospective cohort	3,010 SVBT cycles.	Annotation-free deep-learning embryo scoring system (iDAScore v1.0) using time-lapse sequences	Time-lapse videos (EmbryoScope)	Live birth ≥ 22 weeks per transfer, with total miscarriage and neonatal outcomes as secondary
Borji et al. 2025 [23]	Multi-center	Retrospective cohort	5,000+ IVF cycles	Transformer-based DL + optimization	Clinical data + image features	Live birth prediction
Sadegh-Zadeh et al. 2024 [34]	Iran	Retrospective cohort	4,500+ IVF cycles	Multiple ML paradigms	Comprehensive clinical dataset	Live birth, Clinical pregnancy

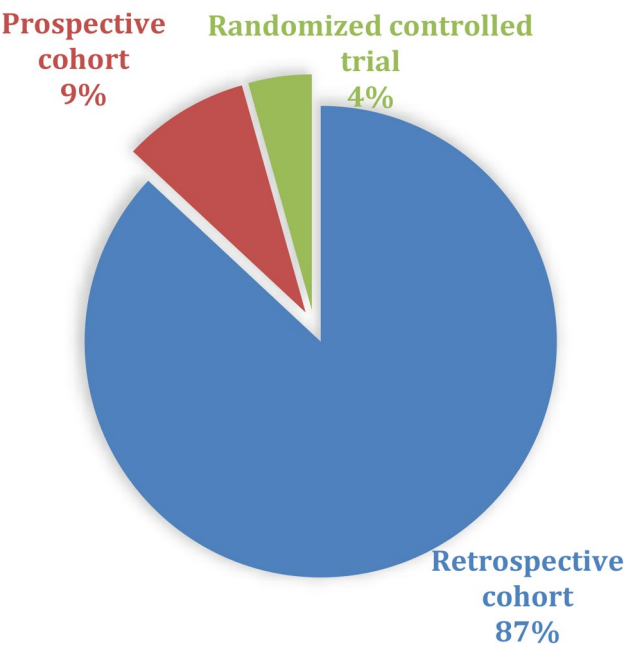


Fig. 2 Proportional representation of study types

prediction using iDAScore in PGT-A cycles. Papamentzelopoulou et al. (2024) [28] directly compared KID-Score D5 (manual morphokinetic) to iDAScore (AI) in 429 blastocysts, reporting AUCs of 0.695 and 0.657 for live birth, respectively. Theilgaard Lassen et al. (2023) [19] analyzed the largest cohort (181,428 embryos) and reported an internal validation accuracy of 77.5% for the iDAScore v2.0 model. Earlier studies, such as Tran et al. (2019) [21], reported an accuracy of 72.1% for predicting fetal heart pregnancy, and Khosravi et al. (2019) [8] reported a lower accuracy of 64.3% for clinical pregnancy prediction across 11 centers.

Static image-based models

In research studies using static images of the blastocyst, the range of accuracy was found to vary from 62% to 69%. Khosravi et al. (2019) [8] found an accuracy of

62.1% using static microscopy images. He et al. (2024) [11] found an accuracy of 68.5% in predicting embryo ploidy and implantation. Enatsu et al. (2022) [7] found an accuracy of 65.2% by combining static images with clinical data. Miyagi et al. (2019) [14] have successfully demonstrated the feasibility of the prediction of live births using static images of the blastocyst in specific age classifications.

Clinical variable-based models

Research studies using only machine learning models without image data have shown accuracy between 76% and 78%. Nguyen et al. (2024) [29] showed that gradient boosting models using clinical and demographic factors from each fertility center significantly outperform conventional models using national registry data in predicting probabilities of live birth per cycle for multiple fertility centers across the United States. Bereczki et al. (2025) [2] reported an accuracy of 78.3% using XGBoost, RF, and SVM algorithms on female preprocedural factors only. Yao et al. (2025) [30] reported an accuracy of 76.8% for center-specific models predicting live birth per cycle, and Sadegh-Zadeh et al. (2024) [34] reported an accuracy of 76.0% using multiple ML models.

Multimodal/ensemble (image + clinical data) models

The models that integrated embryo image data with patient-level clinical variables had the best performance characteristics, with the range of accuracy for each study falling between 74% and 82%. The study by Borji et al. (2025) [23] reported the highest accuracy, i.e., 81.5%, using the transformer-based deep learning model, integrating embryo images with clinical features. Coticchio et al. (2021) [32] reported an accuracy of 73.8% using the hybrid AI model, integrating morphology along with cytoplasmic dynamics using time-lapse imaging.

Comparative performance across model categories

Comparisons of the approaches of the models, conducted directly in the same cohort of samples, also proved to

be informative. In all cases, the need for the inclusion of multimodal approaches, which combined image and clinical information, was demonstrated to have superior performance in terms of accuracy and discrimination in the same populations. Table 2 describes the descriptive ranges of the accuracy and AUC of the respective model categories.

Prospective validation: randomized controlled trial evidence

The only prospective, double-blinded, multicenter randomized controlled study on the use of AI for embryo selection compared with standard morphological analysis was by Illingworth et al. (2024) [31]. This study randomized 1,066 patients, with 533 patients being allocated to the iDAScore algorithm and 533 patients being allocated to standard morphology analysis. The primary outcome was the clinical pregnancy rate. The clinical pregnancy rate was achieved by 46.5% of patients allocated to the iDAScore algorithm, compared with 48.2% of patients allocated to standard analysis. This study was a noninferiority study, but it failed to meet its noninferiority margin, and therefore, the use of an AI algorithm was not shown to be noninferior to standard analysis.

Table 2 Performance characteristics by model type

Model Type	AUC Range (if reported)	Accuracy Range (%)	Number of Contributing Studies	Illustrative Key Studies
Time-lapse based	0.64–0.97	~ 64–78%	11	Huang et al. (2022) [26], Ueno et al. (2022) [33], Theilgaard Lassen et al. (2023) [19]
Static image-based	~ 0.65–0.77	~ 62–69%	4	Khosravi et al. (2019) [8], He et al. (2024) [11], Miyagi et al. (2019) [14]
Clinical variable-based	~ 0.70–0.80	~ 76–78%	4	Nguyen et al. (2024) [29], Bereczki et al. (2025) [2], Yao et al. (2025) [30]
Multimodal (image + clinical)	~ 0.77–0.97	~ 74–82%	4	Borji et al. (2025) [23], Coticchio et al. (2021) [32], Enatsu et al. (2022) [7]

Values represent ranges of study-specific metrics for each model type category and are not summary values. Individual metrics are listed in the text. No meta-analyses or hypothesis tests across studies were conducted

This negative prospective RCT result differs from the highly positive retrospective performance metrics reported for other studies (e.g., Huang et al., 2022; Ueno et al., 2022) [26, 33]. It underscores the necessity of prospective validation of models that might be affected by selection bias or overfitting to specific clinical environments.

Study quality assessment

Study quality was moderate to high across the included trials. Risk of bias assessment using QUADAS-2 revealed variability across the four key domains.

Randomized controlled trial (n = 1)

The Illingworth et al. (2024) RCT demonstrated low risk of bias across all five Cochrane RoB 2.0 domains: randomization process, intervention deviations, missing outcome data, outcome measurement, and selective reporting [4].

Retrospective cohort studies (n = 20)

The majority of the included studies were of a retrospective nature. Some of the limitations of these studies included possible selection biases and possible confounding effects of unknown variables. The methods of validation used were also quite diverse; the majority of the studies used only internal validation methods such as k-fold cross-validation or a holdout test set, whereas only a small number of studies used external validation methods on separate datasets from multiple centers [24].

Risk of bias distribution

- Domain 1 (Patient Selection):* Low risk: 6 studies (26%); Unclear risk: 11 studies (48%); High risk: 6 studies (26%)
- Domain 2 (Index Test):* Low risk: 14 studies (61%); Unclear risk: 7 studies (30%); High risk: 2 studies (9%)
- Domain 3 (Reference Standard):* Low risk: 18 studies (78%); Unclear risk: 4 studies (17%); High risk: 1 study (4%)
- Domain 4 (Flow and Timing):* Low risk: 12 studies (52%); Unclear risk: 9 studies (39%); High risk: 2 studies (9%)

Overall, the greatest area of concern for bias was patient selection (arising from retrospective designs), followed by unclear reporting of flow and timing.

Certainty of evidence assessment

GRADE certainty profile

As shown in Table 3, most of the evidence for predicting live birth came from retrospective cohort studies, and the quality was rated as low certainty according to

Table 3 Outcome-wise summary of study designs and certainty of evidence

Outcome	Number of Studies	Study Design	Over-all Certainty
Live Birth Prediction (all models)	16	Mostly retrospective cohorts + 1 RCT	LOW
Clinical Pregnancy Prediction	15	Mixed designs	MOD-ERATE
AI vs. Standard Morphology	1	RCT	MOD-ERATE
Time-lapse-based AI models	11	Mostly retrospective cohorts + 1 RCT	LOW
Static image-based AI models	4	Retrospective cohorts	LOW
Multimodal (image+clinical) models	4	Retrospective cohorts	MOD-ERATE

Evidence certainty downgraded for risk of bias (retrospective nature), inconsistency, and lack of extensive prospective validation

the modified GRADE criteria. The quality was downgraded because of a serious risk of bias, inconsistency, and imprecision, related to the use of retrospective study design, heterogeneous models, input data, and populations, as well as limited prospective study data.

Discussion

This systematic review considered the existing literature on the application of AI and ML for predicting live birth and IVF outcomes from 23 primary clinical studies. The results of this systematic review indicate that the performance of AI models can be quantified and is promising, but varies widely depending on the model type, data modality, and validation approach. The accuracy of AI models was reported to vary from 60% to 82%, while the AUC varied from 0.64 to 0.97.

Principal findings

Model performance varies by architecture and input data

Multimodal models, which employed both embryo image information and patient-level clinical variables, showed the best performance, achieving accuracies ranging from 74% to 82%, along with the highest AUCs, as seen in Borji et al. (2025) and Huang et al. (2022) [23, 26]. This again highlights the fact that the accurate prediction of live births not only requires the detailed characterization of the embryo itself, but the patient’s systemic reproductive environment must also be modeled. The performance of the clinical variable-based models was average, with accuracies ranging from 76% to 78%, while the static image-based models showed the lowest accuracy, ranging from 62% to 69% [2, 8, 30].

Intended use of models: absolute prediction vs. within-cohort ranking

However, the advantage of multimodal AI models should be viewed in relation to the intended use of the embryo selection procedure. Typically, such AI models are employed to achieve one of two different aims: (1) to predict the absolute probability of a live birth or (2) rank embryos from a specific treatment group to choose the one with the highest probability of success [9]. If the goal is to predict the absolute probability of having a baby across a range of patients, multimodal models can be highly effective since they consider the overall reproductive environment of the patient. If the aim is simply to rank embryos from the same patient, patient characteristics are irrelevant.

Due to the uniform maternal features among all embryos in a cohort, a prediction system using these clinical predictors will automatically generate a within-treatment cohort AUC of 0.5 since all embryos will receive the same score modification due to their identical maternal clinical attributes. This technical disparity makes it essential for most commercially available morphokinetic imaging-based models such as iDAScore, KIDScore, CHLOE, A-Life, and A-IVF to adopt a relative continuous scale that ranks embryos based only on morphokinetics and imaging features, and not on predicting an absolute probability [28; 19]. It is important for clinics to be aware of this disparity when selecting an appropriate AI technology since a model designed for making absolute predictions does not guarantee its superiority in determining which embryo to transplant among a certain group of embryos.

Center-specific models outperform national registries

In addition, research has consistently shown that models trained on local center-specific information have a higher predictive ability compared to models trained on IVF registry information [29, 30]. This suggests that the specific information used in a particular lab has a significant impact on the performance of the models.

Prospective validation reveals performance gap

The only prospective RCT of AI-based embryo selection compared to the standard morphological assessment of embryos by the iDAScore AI model reported clinical pregnancy rates of 46.5% and 48.2%, respectively, and did not demonstrate non-inferiority of the AI-based system. This is in stark contrast to the optimistic performance of the AI-based systems reported in retrospective cohorts (e.g., Ueno et al., 2022) [33]. This is an example of the potential for bias in the retrospective cohorts and the need for prospective RCT validation of the systems.

Comparison with previous evidence

Previous systematic reviews on the application of AI in embryo evaluation have mostly concentrated on various definitions of outcomes such as blastocyst formation, implantation, and clinical pregnancy rather than focusing specifically on live birth outcomes [1, 22]. This review's more specific inclusion criteria of live birth as a primary or secondary outcome or as a validated proxy outcome provide a more accurate picture of the end application of AI.

The results of our review are in line with previous studies that showed time-lapse imaging is generally more effective in terms of diagnostic potential than static imaging due to its ability to capture the morphokinetics of embryonic development [19, 21]. Our review builds on these findings to show that not only are time-lapse imaging and deep learning algorithms such as iDAScore not universally effective in terms of clinical outcomes in prospective randomized settings, but also not seamlessly transferable to real-life settings as previously thought [28, 31].

Clinical practice implications

Evaluating AI-based embryo selection systems is methodologically difficult. Heterogeneity has been noted in the evaluation of AI models for embryo selection systems. In many studies, different objectives are optimized by AI models—embryo ranking within a group or absolute probabilities of live birth—while non-comparable metrics are reported. Reliance on raw accuracy is not advisable when there are large class imbalances (e.g., low rates of live birth), making the Area Under the Curve a more robust primary discrimination metric [9, 10].

Furthermore, selection bias often plagues retrospective studies comparing AI systems and human embryologists, particularly when models are only trained and tested on embryos deemed competent by human embryologists for transfer. There are no standardized reporting formats, making it difficult to compare studies. The newly developed guidance on reporting prediction model performance, denoted as TRIPOD + AI, should be followed by all embryo selection studies moving forward, as it standardizes reporting formats [10].

Biological and mechanistic implications

The superior performance of the multimodal models combining embryo images and clinical variables is biologically highly plausible. Blastocyst morphology and morphokinetics from time-lapse imaging are direct indicators of the embryo's intrinsic developmental potential and genetic quality [26, 33]. On the contrary, clinical variables such as age, ovarian reserve tests, and endometrial factors are indicators of the systemic reproductive environment and

receptivity to which the embryo is exposed during transfer [29, 30].

The deep learning model's focus on the inner cell mass and trophectoderm areas of the embryo in the time-lapse images also aligns with the traditional intuitive approach of an embryologist [11, 19]. However, the mechanistic interpretation of the results is still limited by the fact that deep learning models are a "black box." Currently, only a few studies have used explainability approaches to clearly describe what specific aspects of the embryo's features are being used by the model to make its predictions [7].

Clinical implications

Currently, there is a lack of evidence to support the use of AI by itself to replace traditional assessment by embryologists. In addition, since the only available prospective RCT did not prove the non-inferiority of deep learning algorithms against traditional morphology assessment, AI systems do not seem to possess any clinical superiority on their own [31].

Nevertheless, AI is an excellent complementary tool for decision support. In fact, it has the potential to decrease inter- and intra-observer variability by standardizing the grading of embryos, speed up the evaluation process to reduce the time required for manual evaluation, and act as an electronic quality assurance tool to alert embryos for re-evaluation by the experts [28]. Infertility clinics looking to integrate AI should focus on systems that incorporate various inputs from patients and embryos, have transparent validation data, and place final decision-making authority in the hands of the embryologist.

Limitations

There are a number of notable limitations to be highlighted within this review and the underlying literature. Firstly, 87% of the primary studies reviewed within this article were retrospective cohorts, which are inherently susceptible to issues of selection bias, confounding effects, and spectrum bias—often only considering embryos with definitive outcomes. Furthermore, the methods used for validation were highly variable; the vast majority of studies only used internal validation methods (for instance, k-fold cross-validation), which overestimate the true performance of the model due to over-optimization of certain traits inherent within the underlying cohort.

Thirdly, definitions of outcomes were not uniform. Although live birth is clearly the priority outcome, other surrogate outcomes, such as fetal heartbeat or clinical pregnancy, were used in various studies. Fourthly, the heterogeneity of model types and input data modalities and metrics used (accuracy and/or AUC) precluded a quantitative synthesis and necessitated a qualitative synthesis. Lastly, there is a notable lack of data on certain subpopulations, such

as poor responders, advanced age groups, and causes of infertility.

Future research directions

Essential research priorities

Large-Scale Prospective RCTs: Future studies will need to involve well-designed prospective RCTs with sufficient sample size in which AI-assisted embryo selection is compared to standard morphological assessment, with live birth as the primary outcome measure. Such trials should randomize patients (not embryos), have complete follow-up to live birth, be conducted among diverse populations in different geographic areas, and report results broken down by relevant subgroups.

External Validation Studies. Models that are developed in single-center or limited multicenter cohorts need to be validated externally prospectively in independent cohorts and settings using a wide range of laboratory settings before clinical implementation.

Explainability Research: Studies should use explainability techniques (attention visualization, saliency maps, feature importance analysis) to help understand what features of the embryo are driving the model predictions, which can help validate the logic of the models and make clinicians more confident in the model.

Mechanistic Investigation: Prospective studies should examine the biological mechanisms of predictions of AI. Do high AI scoring embryos have higher rates of chromosomal euploidy (testable through preimplantation genetic testing)? Do they have more favorable metabolomic profiles?

Population-Specific Research Targeted research should be conducted to assess model performance in specific populations (poor responders, advanced maternal age, severe male factor) to determine if models should be recalibrated for specific populations.

Standardization of Reporting: The field should implement standardized reporting guidelines for AI model development and validation studies in reproductive medicine, which would be informed by existing reporting guidelines (TRIPOD for prediction models, CONSORT-AI for RCTs) tailored to IVF applications.

Conclusions

This systematic review of 23 primary clinical studies shows that AI-based predictive models for live birth after IVF have promising though variable diagnostic accuracy with AUCs ranging from 0.64 to 0.97. Multimodal models using both embryo morphokinetics and patient clinical features were consistently superior to single-modality models, emphasizing the importance of assessing both embryo potential and systemic reproductive environment for live birth prediction. Additionally, models trained on center-specific datasets were superior to those trained on national registry datasets.

However, the current body of evidence is largely limited to retrospective cohort studies that are subject to a high risk of selection bias and internal validation overfitting. Notably, the single prospective randomized controlled trial evaluating AI embryo selection compared with standard morphology did not show non-inferiority. Thus, the overall GRADE evidence remains largely low to moderate certainty.

Currently, AI can be considered an adjunct tool for decision support, intended to aid, not supplant, the human embryologist. Future research will need to focus on large-scale, prospective RCTs with live birth outcomes, robust external validation, including in different laboratory settings, the use of multimodal data, and the reporting of studies in accordance with the TRIPOD+ AI statement.

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Prospero registration

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Authors' contributions

RY conceptualized the study, conducted the literature search, performed data extraction and qualitative synthesis, and drafted the manuscript. DB contributed to study selection and methodological appraisal. JR contributed to the critical revision of the manuscript. All authors read and approved the final manuscript.

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Data availability

All data analysed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate

Not applicable. This study is a systematic review based on previously published literature and does not involve human participants, human data, human tissue, or animals.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data.

Competing interests

The authors declare no competing interests.

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